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grep

Definition: `grep` filters through text and shows lines that match.

Usage / Formula: `grep [options] pattern file`

Example 1: `grep Honda cars.csv`

Example 2: `grep -i "^honda" cars.csv`

- `grep` → command used to search text
 - `-i` → ignores uppercase/lowercase differences
 - `"^honda"` → `^` means "start of the line", so it matches lines that start with Honda
 - `cars.csv` → file being searched
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awk

Definition: `awk` is used to work with columns in a file.(Better cut)

Usage / Formula: `awk 'Field { action }' file`

Example 1: `awk '{print $1}' cars.csv`

- Prints the first column of each line

Example 2: `awk -F"," '{print $1, $3}' cars.csv`

- `awk` → command
 - `-F","` → sets the field separator to a comma
 - `{print $1, $3}` → prints column 1 and column 3
 - `cars.csv` → file
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sed

Definition: `sed` edits text.

Usage / Formula: `sed 's/pattern/replacement/' file`

Example 1: `sed 's/;/,/g' cars.csv`

- Replaces semicolons with commas

Example 2: `sed 's/Honda/Toyota/g' cars.csv`

- `sed` → command
- `s` → substitute command
- `Honda` → text to change

- `Toyota` → replacement text
 - `g` → replace all matches on each line
 - `cars.csv` → file being edited
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Pipe |

Definition: | sends the output of one command into another command.

Example 1: `ls | grep txt`

- Sends file list to `grep` to show only `.txt` files

Example 2: `grep Honda cars.csv | sed 's/;/,/g'`

- `grep Honda cars.csv` → finds Honda lines
- `|` → passes result to next command
- `sed 's/;/,/g'` → replaces semicolons with commas

redirecting Output to a File >

Definition: > saves command output to a file and overwrites it(deletes what was in it and put the new output).

Example 1: `ls > files.txt`

Example 2: `grep 'Honda' cars.csv > honda.csv`

- `grep Honda cars.csv` → finds Honda
 - `>` → redirects output
 - `honda.csv` → file created or overwritten
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Append Output to a File >>

Definition: >> adds output to the end of a file without deleting the data that was already in it.

Example 1 (Simple): `date >> log.txt`

Example 2 (More in-depth): `grep 'Honda' cars.csv >> honda.csv`

- `grep Honda cars.csv` → gets Honda lines
- `>>` → adds output
- `honda.csv` → file being added to